

CHELYABINSK MEZHDUNARODNYY, Russian Federation

WMO: 286420

Lat: 55.306N

Lon: 61.503E

Elev: 769

StdP: 14.29

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-15.1	-26.2	1.3	-20.1	-20.8	1.8	-14.5	23.1	17.3	20.9	18.1	5.7	330	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	87.5	66.6	84.0	65.6	80.6	64.4	69.8	81.3	68.0	79.1	66.4	76.9	9.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	98.9	73.6	64.3	92.8	72.6	62.6	87.2	70.8	34.1	81.5	32.8	79.1	31.4	76.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.8	20.0	17.8	DB	-26.1	92.0	5.8	3.5	-30.3	94.5	-33.7	96.6	-37.0	98.6	-41.3	101.2
(5)				WB	-26.3	72.7	5.7	2.5	-30.4	74.5	-33.8	75.9	-37.0	77.3	-41.1	79.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	6.9	10.5	23.9	41.5	54.7	63.3	66.2	62.8	52.4	39.7	23.1	12.3
	DBStd	23.31	12.13	12.10	8.85	9.83	9.31	7.84	6.71	7.60	8.15	9.03	12.47	12.25
	HDD50	5987	1336	1107	808	289	60	6	0	2	65	336	808	1170
	HDD65	10036	1801	1527	1273	707	339	123	68	135	386	784	1258	1635
	CDD50	1698	0	0	0	33	205	404	503	400	136	17	0	0
	CDD65	272	0	0	0	1	20	71	105	68	7	0	0	0
	CDH74	2780	0	0	0	17	264	724	1009	667	99	0	0	0
	CDH80	819	0	0	0	2	53	219	304	218	23	0	0	0
Wind	WSAvg	9.1	8.0	8.3	10.0	10.4	10.1	9.3	8.2	8.4	9.0	9.6	9.2	8.5
Precipitation	PrecAvg	17.4	0.7	0.6	0.6	0.9	1.7	2.3	3.5	2.2	1.4	1.3	1.0	0.9
	PrecMax	25.5	2.8	2.0	2.1	2.5	3.6	4.5	7.2	5.9	5.6	3.0	2.0	1.8
	PrecMin	9.4	0.0	0.0	0.0	0.0	0.0	0.2	1.0	0.3	0.0	0.2	0.1	0.0
	PrecStd	3.3	0.6	0.4	0.4	0.7	0.9	1.1	1.6	1.3	1.1	0.8	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	39.0	50.0	76.6	85.6	90.3	91.6	91.1	83.8	67.8	48.6	38.9
		MCWB	32.8	34.0	42.2	56.4	63.2	65.7	68.5	68.8	61.7	52.6	43.2	34.9
	2%	DB	31.3	35.3	42.9	68.4	80.7	86.0	86.4	85.9	76.6	62.2	44.3	34.3
		MCWB	28.9	31.6	36.8	52.0	59.9	65.7	67.1	67.0	59.0	50.5	40.3	31.4
	5%	DB	28.1	31.7	39.1	62.9	76.8	82.2	82.8	81.0	70.4	57.3	40.9	32.1
		MCWB	26.1	28.8	34.4	49.5	58.4	64.6	66.5	65.8	57.4	48.0	37.5	29.8
	10%	DB	24.7	28.1	36.0	57.4	71.8	78.2	80.0	76.9	66.1	52.9	37.6	28.8
		MCWB	23.0	25.4	32.2	46.5	56.3	63.0	65.6	63.6	55.1	46.3	34.6	26.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	35.0	42.3	57.3	65.5	70.6	71.9	71.7	64.4	54.8	43.7	35.5
		MCDB	35.0	37.8	49.1	74.0	79.5	82.1	82.4	85.4	77.3	63.6	47.0	37.6
	2%	WB	28.9	31.5	37.7	53.1	62.1	68.3	69.9	69.2	60.9	51.7	40.5	32.1
		MCDB	30.7	34.6	42.0	66.1	75.8	80.6	81.1	80.9	71.7	59.3	43.4	33.8
	5%	WB	26.0	28.7	34.9	50.4	59.8	66.5	68.4	67.0	58.6	49.1	37.5	30.0
		MCDB	27.7	31.4	37.8	61.3	72.7	77.9	78.8	77.9	68.6	55.9	40.1	32.1
	10%	WB	22.8	25.5	32.8	47.5	57.4	64.5	66.9	65.0	56.3	46.6	34.9	27.0
		MCDB	24.4	27.8	35.7	56.4	70.1	74.8	76.7	74.6	64.6	51.9	37.1	28.8
Mean Daily Temperature Range	5% DB	MDBR	13.8	15.9	15.5	18.4	21.3	19.7	19.1	18.3	17.6	13.3	11.7	12.8
		MCDBR	13.8	15.2	16.6	26.6	27.2	25.1	24.4	24.9	24.8	21.2	14.0	14.1
		MCWBR	12.5	13.0	12.3	14.3	12.4	10.4	9.9	10.1	11.8	11.9	11.0	12.2
	5% WB	MCDBR	13.2	14.3	15.1	24.6	24.2	22.2	21.2	22.4	22.9	19.0	12.6	14.0
		MCWBR	12.2	12.6	11.8	14.1	12.6	10.7	10.1	10.4	12.2	11.7	10.6	12.3
Clear-Sky Solar Irradiance	taub	0.282	0.311	0.370	0.408	0.407	0.404	0.431	0.444	0.384	0.354	0.309	0.280	
	taud	2.069	2.047	1.987	2.125	2.233	2.312	2.227	2.163	2.296	2.255	2.161	2.044	
	Ebn at Noon	193	233	245	256	265	267	256	242	243	218	182	161	
	Edh at Noon	21	31	42	43	41	38	41	41	31	25	19	17	
All-Sky Solar Radiation	RadAvg	269	608	1083	1390	1649	1792	1739	1357	916	460	233	166	
	RadStd	25	34	75	144	159	161	124	134	72	80	26	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page